



Helmut Chaparro

Electronics Engineer

Electronics engineer — Data Analyst — Full-stack Developer — Roboticist — Entrepreneur

CONTACT INFORMATION

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LANGUAGE PROFICIENCY

- Native Spanish speaker
- English B2 level
- French B2 level
- Mandarin Chinese HSK2 level
- German A1 Level

TECHNICAL PROFICIENCIES

- Altium designer & KiCAD software.
- Fusion 360.
- Blender Studio.
- Adobe Suite.
- Cursor
- Cisco Packet Tracer
- Office suite.
- Matlab - Octave.
- SQL server.
- XILINX - INTEL QUARTUS
- R software
- ROS2
- GAZEBO
- STM32 CUBE Tools
- WeBots cyberbotics
- QT Creator & QT design studio
- Ecostruxure Machine Expert
- LTSPICE
- ARDUINO IDE & ARDUINO APPLAB
- ESP32 IDF
- Node.js
- Tensorflow
- OpenCV
- RTOS
- MCP Server
- Git/Github

PROFESSIONAL PROFILE

Electronics Engineer, Full-Stack Developer, Data Analyst, Roboticist, and Entrepreneur with over six years of experience leading engineering, software, embedded systems, robotics, IoT, data analysis, and STEM education projects. Skilled in .NET, Python, C/C++, C#, JavaScript, SQL, STM32, ESP32, PCB design, FPGA development, automation, and full-stack development.

Co-founder and General Manager of DevKit Electronics S.A.S., leading strategy, operations, product development, and the creation of hands-on engineering education solutions for schools, universities, and professional training programs. Experienced in building scalable educational technologies, robotics kits, laboratory systems, software platforms, and technical curricula.

Proven leader in multidisciplinary environments, with experience in corporate, academic, and public-sector projects, IEEE leadership roles, and national recognition as a hackathon winner for SAT-ViRe, a low-cost landslide early warning system for vulnerable communities in Colombia. Focused on developing innovative, scalable, and socially impactful technology solutions.

ACADEMIC BACKGROUND

● ELECTRONICS ENGINEER

Universidad Distrital Francisco José de Caldas
2020-2027

PROFESSIONAL CREDENTIALS

● CERTIFIED IN PROFESSIONAL PCB DESIGN AND ASSEMBLY

Altium Designer Software - Universidad Distrital Francisco José de Caldas
2023

● DATA ENGINEERING

Deeplearning.ai
2026

● ALTIUM GLOBAL SCHOLARSHIP PROGRAM

Altium Designer Software
2026

● TELEDETECCIÓN Y DRONES

ESRI Colombia
2026

● JAVASCRIPT BACKEND DEVELOPER

IBM
2026

● AI DEVELOPER

IBM
2026

● DATA SCIENCE PROFESSIONAL

IBM
2026

● MICROSOFT AI & ML ENGINEERING

Microsoft
2026

● RAG AND AGENTIC AI

IBM
2026

● GOOGLE ADVANCED DATA ANALYTICS

Google
2026

CORE TECHNICAL EXPERTISE

- Embedded systems and microcontroller programming
- C, C++, C#, Python, JavaScript and Node.js development
- Full-stack development with .NET, REST APIs and backend services
- FPGA design, digital logic and hardware description using VHDL and Verilog
- STM32, ESP32, Arduino, Raspberry Pi and SoC-based development
- Professional PCB design, multilayer routing, circuit simulation and electronic prototyping
- Sensor integration, IoT architectures and real-time data acquisition systems
- Robotics, automation, control systems and mechatronic prototyping
- Data analysis, statistical computing and predictive modeling with Python and R
- SQL Server, database management, data modeling and query optimization
- Advanced Excel, Power BI, dashboards and automated reporting workflows
- MATLAB/Octave, LTspice, KiCad, Altium, Proteus, Fusion 360 and Blender
- 3D modeling, product design, rapid prototyping and additive manufacturing
- Git/GitHub, Azure DevOps, version control and technical documentation
- UX/UI design, educational software and interactive learning platform development
- Development of STEM, robotics and engineering education solutions
- Technical project coordination, product roadmapping and multidisciplinary team leadership
- Cybersecurity fundamentals, cloud-based workflows and MCP server integration

PROFESSIONAL PROFILE

- Strategic leadership
- Effective bilingual communication
- Creative problem-solving
- Analytical and critical thinking
- Management of multidisciplinary teams
- Interdisciplinary collaboration
- Innovation and disruptive thinking
- Data-driven decision-making
- Adaptability and flexibility
- Social and ethical commitment
- Professional resilience

PROFESSIONAL EXPERIENCE

● JUNIOR FULLSTACK DEVELOPER

Tranexco. Full-time - Remote
May 2024 - April 2025

- I designed .NET solutions for customer management, covering customs processes, imports, DIAN compliance, logistics, and integrations.
- I developed robust solutions for high-volume clients, optimizing international trade workflows and ensuring scalability, security, and performance.

● ROBOTICS AND ELECTRONICS TEACHER

COMPENSAR - Universidad COMPENSAR - in person

July 2025 - December 2025

- Leadership and implementation of STEAM initiatives focused on reducing the educational gap in District Educational Institutions (IEDs), promoting science and technology at the school level.
- Design and implementation of specialized courses in robotics, automation, and embedded systems with AI, successfully integrating them into UCOMPENSAR's curricular projects.

● SOFTWARE DEVELOPER, PROGRAMMER AND EDUCATIONAL ROBOTICS CONSULTANT

Arisoft Seguridad Vial S.A.S - Full-time - hybrid

June 2023 - November 2024

- I managed educational robotics projects for commercialization to district and educational institutions.
- My responsibilities included programming educational robots, developing electronic circuits, and providing on-the-job consulting.

● CONSULTING FIRM SPECIALIZING IN ENGINEERING AND EDUCATIONAL ROBOTICS SOLUTIONS

Computadores para educar - Ministerio de educación

March 2021 - November 2023 - Full-time - Hybrid

- I led the complete design and development cycle (from conceptualization to implementation) of innovative mechatronic prototypes and projects, implementing STEAM methodology.
- I applied electronic design (PCB), 3D modeling, robotics, programming (Python, C/C++), and UI development to solutions focused on education and vulnerable populations.

● POSITIONS AT THE PRESIDENCY AND STAFF OF THE IEEE UNIVERSIDAD DISTRITAL

Institute of Electrical and Electronics Engineers universidad Distrital (IEEE UD)

June 2022 - Present 2026. - Part-time - Hybrid.

- I implemented and managed presidential positions in the IEEE branch at the Universidad Distrital Francisco José de Caldas, as well as presidential positions in the Robotics and Automation Society, the Engineering in medicine and Biology Society, and the Humanitarian Technologies chapter (IEEE UD, RAS UD, EMBS UD, and SIGHT UD presidencies). I currently hold positions as president of the student chapters of RAS, SIGHT, and CIS.

● FOUNDER

Devkit Electronics S.A.S

January 2024 - Present 2026. - Part-time - Hybrid.

- Founder and led DevKit Electronics S.A.S., overseeing business strategy, operations, product direction, and the development of hands-on STEM and engineering education solutions for schools, universities, and professional training programs.
- Designed and delivered end-to-end educational engineering products, including curricula, multilayer PCB hardware, embedded firmware, educational software, web platforms, robotics kits, and laboratory systems, while leading multidisciplinary teams across electronics, software, and pedagogy.

● NATIONAL HACKATON WINNER

PNUD/UNGRD Colombia

May 2026 - Winning Pitch - In Person Category

- Awarded winner in the Early Warning Systems category for landslides and slope failures. Developed SAT-ViRe, a low-cost Early Warning System designed to detect risk levels and support landslide prediction in rural and vulnerable areas of Villarica, Tolima, and parts of Soacha, Colombia. The system alerts both disaster risk management organizations and affected communities, strengthening risk communication, data collection, prevention efforts, and safer decision-making for housing and settlement planning.